

Social Media Sentiment Analysis Dashboard

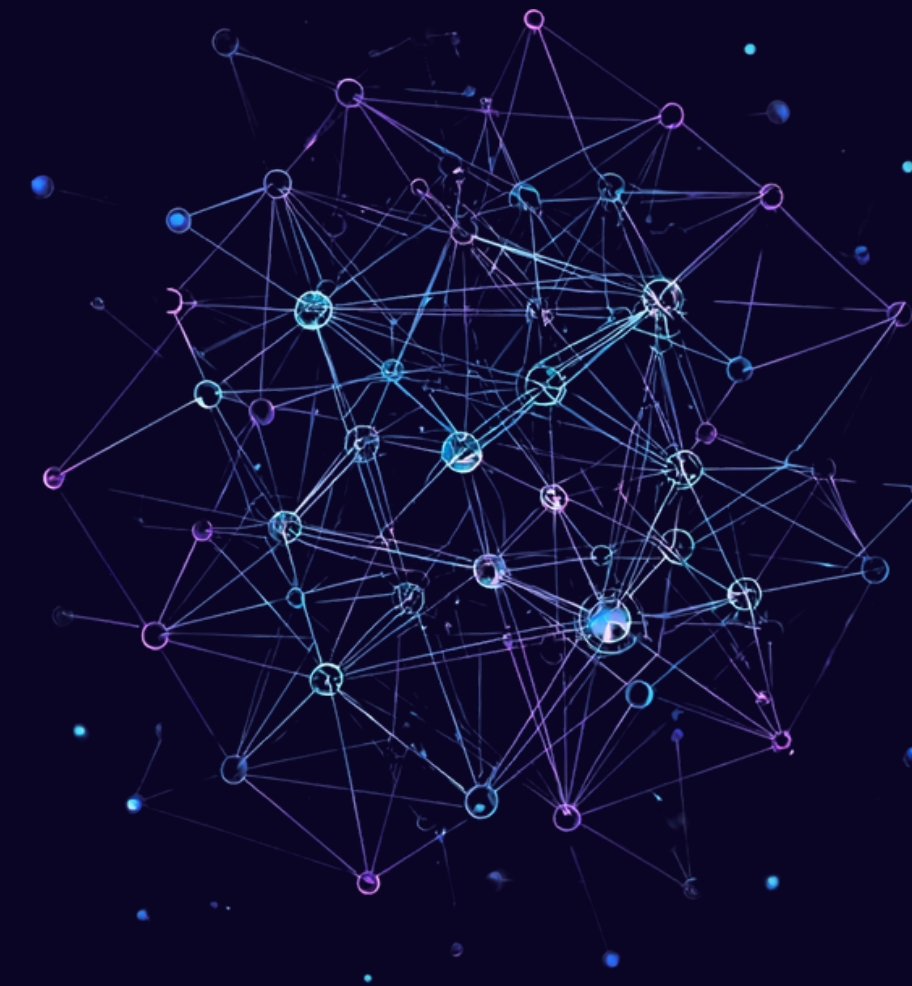
Power BI Project



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Problem Statement & Objectives

Real-time understanding of public opinion
Brand perception and reputation risk identification
Enabling data-driven decision making
Analyze sentiment patterns across social media
platforms
Identify temporal trends and compare platform
behaviors
Create interactive dashboard for stakeholder decisions



Dataset Overview

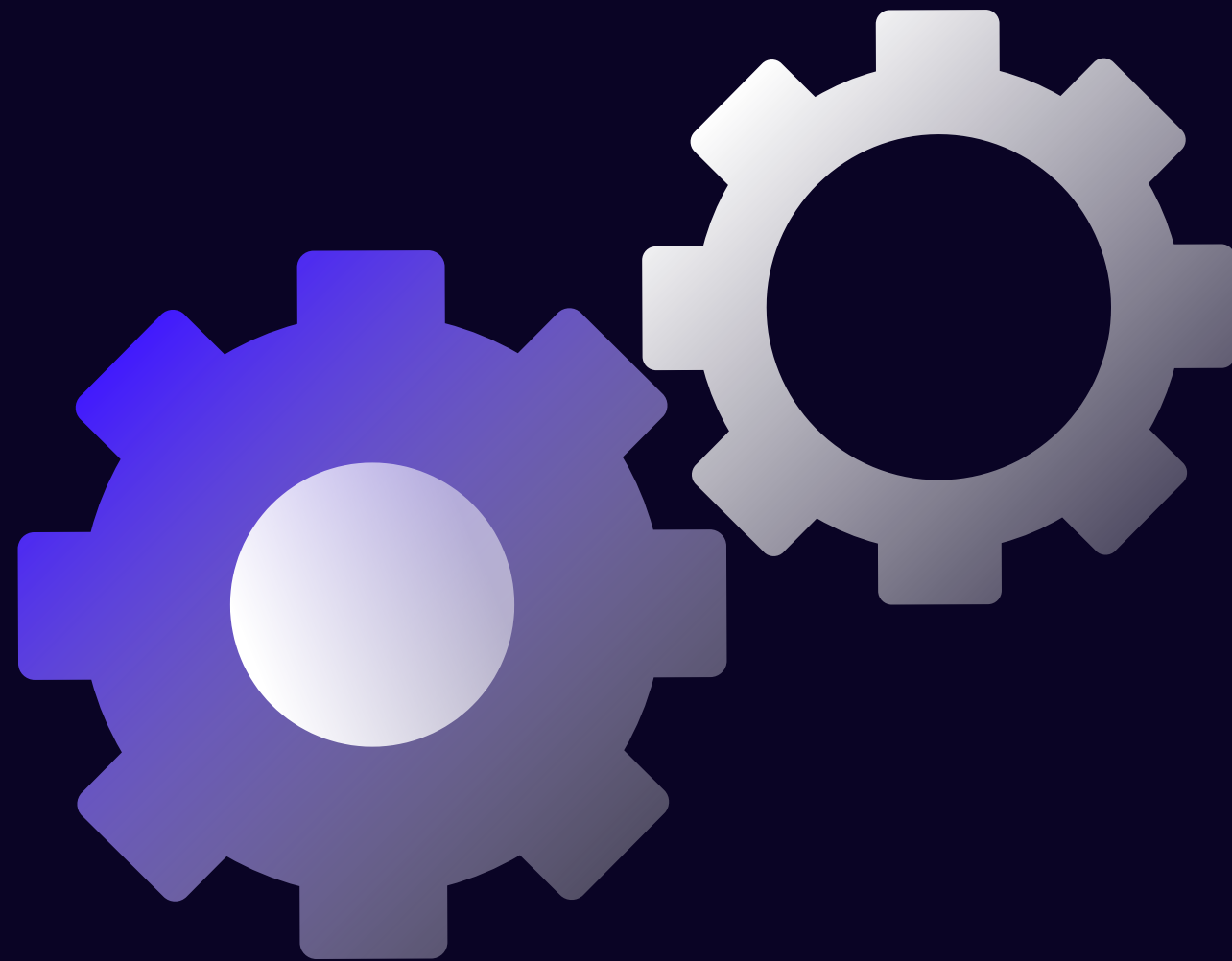
~500 Social Media Posts

Twitter · Facebook · Instagram

Sentiment: Positive / Neutral / Negative

Temporal: Year, Month, Day, Time

Text Content & Platform Labels



Data Preparation & Methodology

Data Cleaning: Missing values,
duplicates, text formatting

Data Transformation: Time-of-day
categories, sentiment scores

Sentiment Categorization: Positive,
Neutral, Negative labels

DAX Calculations: KPIs, sentiment %,
aggregations

Dashboard Development: Visuals, filters,
drill-downs

Dashboard Overview

Key Performance Indicators

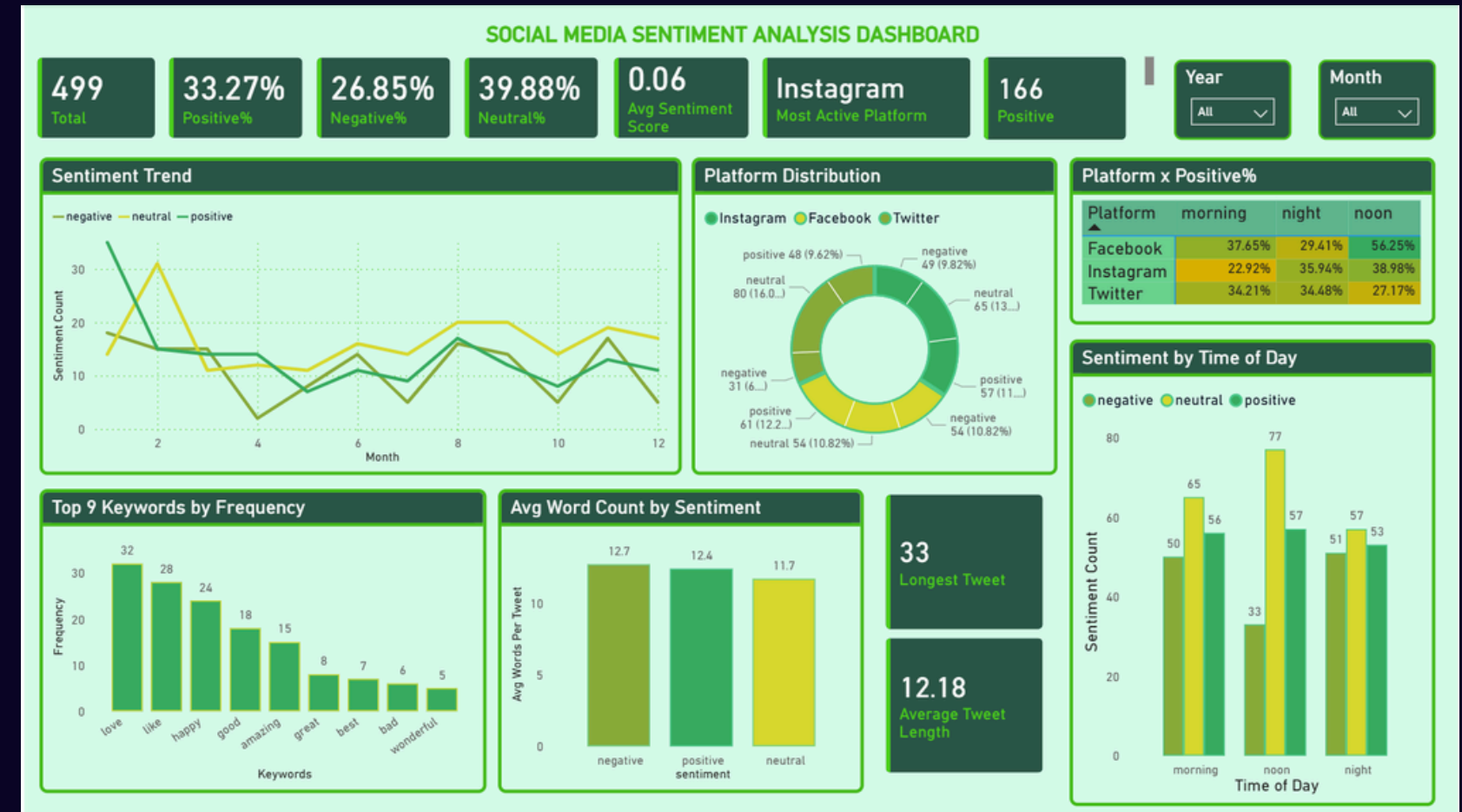
Total Posts: 499

- Positive Sentiment: 33.27%
- Neutral Sentiment: 26.85%
- Negative Sentiment: 39.88%
- Avg. Sentiment Score: 0.06

Dashboard Layout Highlights

Top: KPI cards

- Left: Sentiment trends & keyword analysis
- Center: Platform distribution & sentiment breakdown
- Right: Time-based & comparative sentiment analysis



Interactive Features

Filters by platform & time period

- Sentiment drill-down capabilities
- Cross-filtering across visuals
- Trend line, donut chart, bar charts & keyword frequency visuals



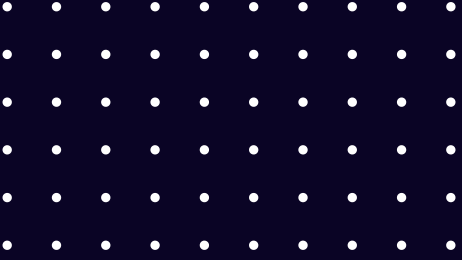
Sentiment Analysis Insights

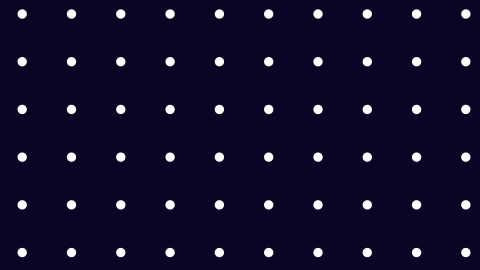
Neutral: 39.88%

Positive: 33.27%

Negative: 26.85%

Avg Score: 0.06





Platform Analysis

Platform Distribution

Instagram is the most active platform with 166 posts, followed by Twitter and Facebook with comparable representation. Each platform contributes meaningfully to the overall dataset of 499 posts.

Key Insights

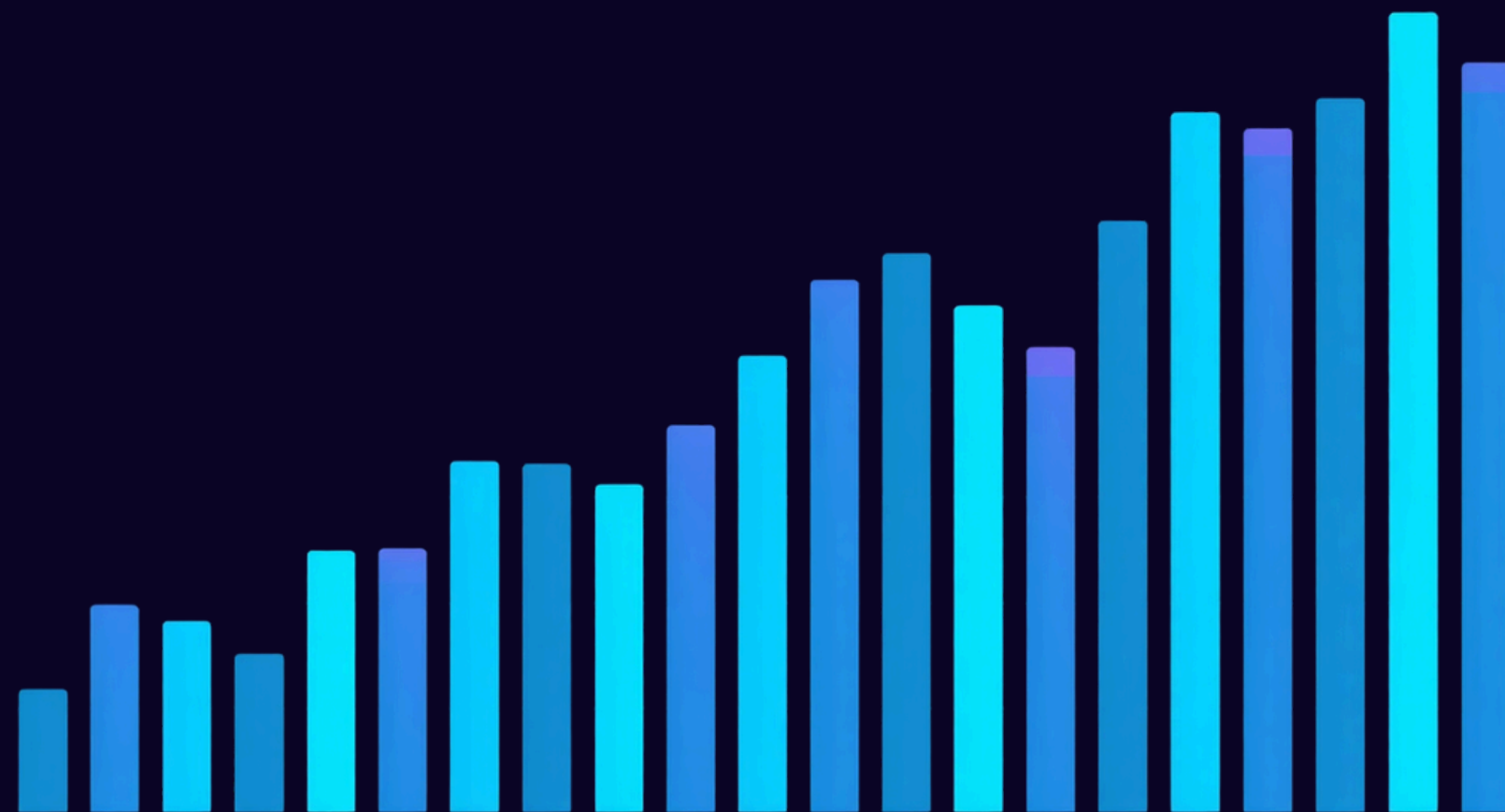
Instagram leads in activity and positive engagement potential. Each platform reflects distinct audience behaviors and communication styles, producing unique sentiment signatures across the dataset.

Sentiment by Platform

Stacked bar charts reveal unique sentiment patterns per platform. Negative sentiment is consistently present across all three, while positive and neutral ratios vary, highlighting platform-specific audience dynamics.

Strategic Implications

Tailor content and engagement strategies per platform. Allocate resources based on platform activity levels and sentiment trends. Deploy platform-specific approaches to improve overall sentiment performance.

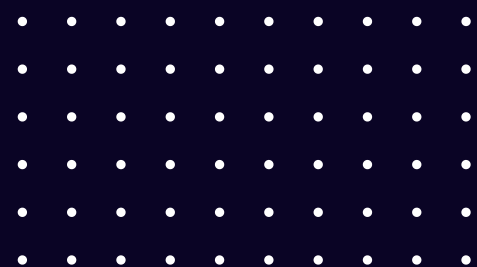
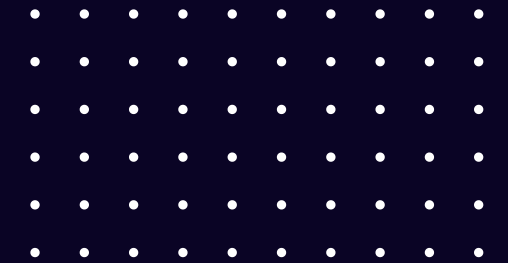


Behavioral & Trend Analysis

Sentiment trends show monthly fluctuations across platforms
Morning & evening peaks reveal distinct user mood patterns
Platform activity heatmap identifies optimal posting windows
Negative feedback spikes linked to specific time periods

Text Analytics Insights

Top Positive Keywords: Themes to amplify in marketing campaigns
Top Negative Keywords: Pain points and early crisis warning signals
Negative posts longest (~33 words) — detailed complaints and feedback
Neutral posts shortest (~12.18 words) — brief and factual content
Positive posts moderate length — concise praise and endorsements
Keyword monitoring enables targeted and proactive brand response
Text length serves as a supplementary sentiment strength indicator



Conclusion & Recommendations



Key Findings: Neutral sentiment dominates (39.88%)

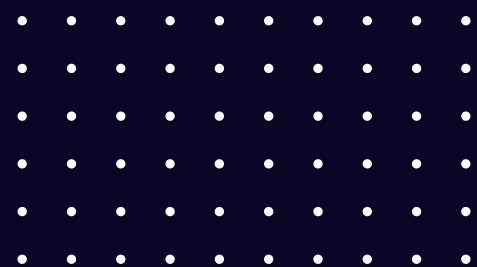
- Instagram leads in activity
- Sentiment varies by time and platform
- Negative posts are longer and more detailed

Recommendations: Investigate drivers of negative sentiment

- Develop platform-specific content plans
- Optimize posting schedules
- Leverage positive keywords in marketing
- Implement real-time sentiment monitoring

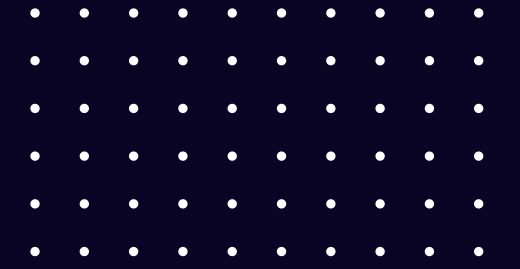
Future Enhancements: Add engagement metrics

- Expand to more platforms with real-time updates
- Incorporate predictive sentiment models and CRM integration





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Thank You!

Questions & Discussion

